mirror-descent-lean: Formal Proofs of Mirror Descent and Bregman Divergence Convergence in Lean 4

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Abstract

`mirror-descent-lean` is a Lean 4 / Mathlib library formalising the core convergence argument for mirror descent with Bregman divergences. The library works over a real inner product space, packages a mirror map with its gradient and strong convexity modulus, defines the Bregman divergence generated by that mirror map, states the mirror descent update through the Bregman projection optimality condition, and proves the standard averaged `O(1/sqrt K)` convergence rate for bounded subgradients. The formal development contains zero `sorry`, zero `admit`, and uses standard Lean/Mathlib axioms only. Its significance is twofold: it gives a machine-checked account of one of the central first-order methods in convex optimisation, and it supplies a reusable Lean artifact for future formal work on optimisation geometry, learning theory, and AI safety.

1. Introduction

Mirror descent is the geometry-aware generalisation of projected gradient descent. In Euclidean gradient descent, progress is measured by squared Euclidean distance. In mirror descent, the distance measure is replaced by a Bregman divergence generated by a strongly convex mirror map. This lets the algorithm adapt its geometry to the feasible region. On Euclidean domains, the quadratic mirror map recovers projected gradient descent. On the probability simplex, the negative entropy mirror map gives exponentiated gradient and multiplicative weights.

The mathematical spine is classical. Let `E` be a real inner product space, let `C` be a convex feasible set, and let `φ : E -> ℝ` be a differentiable, strongly convex mirror map with gradient `gradφ` and strong convexity modulus `σ > 0`. The Bregman divergence is

```
D_φ(x, y) = φ(x) - φ(y) - <gradφ(y), x - y>.
```

For an objective `f`, a subgradient `g_k` at `x_k`, and a positive step size `α_k`, mirror descent chooses `x_{k+1}` by minimising the linearised objective plus the Bregman penalty. Rather than formalising this update as an argmin operator, the library uses the variational inequality that characterises the Bregman projection:

```
forall u in C:
  <gradφ(x_{k+1}) - gradφ(x_k) + α_k g_k, u - x_{k+1}> >= 0.
```

This formulation is important. In the unconstrained case it reduces to the familiar dual-space update

```
gradφ(x_{k+1}) = gradφ(x_k) - α_k g_k.
```

In the constrained case it also captures projection back onto `C`. A single theorem therefore covers both settings.

`mirror-descent-lean` formalises the standard proof that the average regret after `K` steps is bounded by

```
D_φ(x*, x_0) / (K α) + α G^2 / (2σ)
```

for constant step size α and subgradient norm bound `G`. With the optimal constant choice

```
α = sqrt(2σ D_0 / (G^2 K)),
```

the library proves the usual `O(1/sqrt K)` rate

```
average regret <= G * sqrt(2 D_0 / (σ K)).
```

The result is not a new optimisation theorem. It is a precise, importable Lean 4 formalisation of the standard mirror descent convergence argument under explicit hypotheses.

2. Library Overview

The project is organised into four implementation modules plus a root import file:

- `MirrorDescent/Defs.lean` defines `BregmanDiv`, `IsSubgradientAt`, first-order convexity conditions, the `MirrorMap` structure, and `MirrorDescentStep`.
- `MirrorDescent/BregmanDivergence.lean` proves non-negativity, the zero iff equal characterisation under strict convexity, and the three-point identity.
- `MirrorDescent/MirrorStep.lean` proves the strong convexity lower bound, the gradient bound from update optimality, Young's inequality for inner products, the core inner-product estimate, and the per-step mirror descent bound.
- `MirrorDescent/Convergence.lean` defines `MirrorDescentTrajectory` and proves the telescoping sum, the constant-step averaged regret bound, and the `O(1/sqrt K)` rate.
- `MirrorDescent.lean` is the root module importing the library.

The project depends on:

- Lean `v4.28.0`
- Mathlib `v4.28.0`

The formal development contains zero `sorry`, zero `admit`, and introduces no project-specific axioms. It is written against Lean 4 and Mathlib, using standard Lean/Mathlib axioms only.

The formal setting is a real inner product space:

```
variable {E : Type*} [NormedAddCommGroup E] [InnerProductSpace ℝ E]
```

The main definition is the Bregman divergence:

```
def BregmanDiv (φ : E -> ℝ) (gradφ : E -> E) (x y : E) : ℝ :=
  φ x - φ y - inner ℝ (gradφ y) (x - y)
```

The library uses first-order convexity hypotheses rather than deriving them from a separate convex-analysis hierarchy. This keeps the proof surface focused on the mirror descent argument itself. A `MirrorMap` packages the convex domain, the mirror function, its gradient, strict first-order convexity, and first-order strong convexity with modulus `σ`.

A single mirror descent step packages the objective, current iterate, next iterate, optimal point, subgradient, step size, membership hypotheses, subgradient condition, and update optimality condition. The trajectory structure then repeats this data over `K` steps and adds a uniform subgradient norm bound and optimality of `xstar`.

The repository is available at:

https://github.com/velvetmonkey/mirror-descent-lean

3. Theorem Inventory

The source contains thirteen headline results, organised into three layers: Bregman divergence facts, per-step mirror descent analysis, and convergence.

Layer 1 - Bregman Divergence

1. `BregmanDiv.nonneg` — The Bregman divergence is non-negative under the first-order convexity condition:

```
D_φ(x, y) >= 0.
```

This is the basic geometric fact that the tangent hyperplane at `y` lies below the value at `x`.

2. `BregmanDiv.eq_of_div_eq_zero` — Under strict first-order convexity, zero Bregman divergence forces equality:

```
D_φ(x, y) = 0 -> x = y.
```

3. `BregmanDiv.div_eq_zero_of_eq` — The divergence from a point to itself is zero:

```
D_φ(x, x) = 0.
```

4. `BregmanDiv.nonneg_iff_eq` — Non-negativity and equality are combined into the standard zero-characterisation:

```
D_φ(x, y) >= 0, with equality iff x = y.
```

5. `BregmanDiv.three_point` — The Bregman three-point identity:

```
D_φ(x, z) =
  D_φ(x, y) + D_φ(y, z) + <gradφ(y) - gradφ(z), x - y>.
```

This identity is purely algebraic, but it is the calculation that makes the convergence proof telescope.

Layer 2 - Per-Step Analysis

6. `bregman_strong_convex_lower_bound` — Strong convexity lower-bounds the Bregman divergence by squared norm:

```
D_φ(x, y) >= σ / 2 * ||x - y||^2.
```

7. `mirror_update_gradient_bound` — The Bregman projection optimality condition implies the gradient bound needed in the one-step estimate:

```
α_k <g_k, x_{k+1} - u>
  <= <gradφ(x_{k+1}) - gradφ(x_k), u - x_{k+1}>.
```

8. `inner_le_young` — Young's inequality for inner products:

```
a <g, v> <= a^2 / (2σ) * ||g||^2 + σ / 2 * ||v||^2.
```

9. `mirror_update_inner_bound` — The core estimate combines the update condition, the three-point identity, Young's inequality, and strong convexity:

```
α_k <g_k, x_k - u>
  <= D_φ(u, x_k) - D_φ(u, x_{k+1})
     + α_k^2 / (2σ) * ||g_k||^2.
```

10. `mirror_descent_step_bound` — The one-step regret bound:

```
f(x_k) - f(x)
  <= (D_φ(x, x_k) - D_φ(x, x_{k+1})) / α_k
     + α_k / (2σ) * ||g_k||^2.
```

This is the local inequality from which the global convergence rate is obtained.

Layer 3 - Convergence

11. `telescoping_sum` — Summing the per-step bounds over `k < K` gives a finite summed inequality. This is the bridge from one-step analysis to averaged regret.
12. `convergence_constant_step` — For a constant step size α, bounded subgradient norms `||g_k|| <= G`, and initial divergence `D_0 = D_φ(x*, x_0)`, the averaged regret satisfies

```
(1 / K) * sum_k (f(x_k) - f(x*))
  <= D_0 / (K α) + α G^2 / (2σ).
```

13. `convergence_rate` — With

```
α = sqrt(2σ D_0 / (G^2 K)),
```

the averaged regret satisfies the standard mirror descent rate

```
(1 / K) * sum_k (f(x_k) - f(x*))
  <= G * sqrt(2 D_0 / (σ K)).
```

This is the library's final `O(1/sqrt K)` convergence theorem.

4. Key Technical Highlights

Three-Point Identity

The algebraic core of the library is `BregmanDiv.three_point`. Informally, the proof expands the three Bregman divergences and cancels the function values of φ. What remains is a relation between the two gradients and the displacement vector. In ordinary mathematical writing this calculation is short, but in a proof assistant it is exactly the kind of identity that must be stated in the right orientation for the later telescoping argument.

The identity lets the proof compare `D_φ(x, x_k)` and `D_φ(x, x_{k+1})`. In the one-step analysis, the difference between these two divergences exposes a cross term involving

```
<gradφ(x_{k+1}) - gradφ(x_k), x - x_{k+1}>.
```

The mirror update optimality condition controls that cross term. The subgradient inequality then converts an inner product with `g_k` into objective regret. This is the mechanism by which the Bregman geometry turns into an optimisation bound.

The important feature is not merely that the identity is true. It is that it has the exact shape needed for telescoping: after summing over `k`, the adjacent Bregman terms cancel and leave only the initial divergence and a non-negative terminal term.

Bregman Projection Optimality

The update is expressed through the Bregman projection optimality condition rather than through a raw gradient equation. This is a more general statement of the algorithm. It says that the chosen next point `x_{k+1}` satisfies the variational inequality associated with minimising

```
α_k <g_k, u> + D_φ(u, x_k)
```

over the feasible set `C`.

When there are no active constraints, the condition collapses to the dual-space update

```
gradφ(x_{k+1}) = gradφ(x_k) - α_k g_k.
```

When constraints are present, the same inequality describes the Bregman projection back to `C`. This uniform treatment avoids splitting the library into separate constrained and unconstrained developments. It also matches the way mirror descent is normally used in optimisation: the geometry of the domain is part of the method, not an afterthought added to an unconstrained gradient step.

In the Lean proof, this condition supplies exactly the inequality needed to control the term involving `gradφ(x_{k+1}) - gradφ(x_k)`. The formal result is deliberately stated as an inequality, because that is the reusable form for constrained optimisation.

Strong Convexity Lower Bound

Strong convexity connects the Bregman geometry back to the ambient norm geometry. The formal hypothesis says that for all feasible `x` and `y`,

```
φ(x) >= φ(y) + <gradφ(y), x - y> + σ / 2 * ||x - y||^2.
```

Rearranging this statement gives

```
D_φ(x, y) >= σ / 2 * ||x - y||^2.
```

This lower bound is used in the one-step estimate to absorb the norm-square term produced by Young's inequality. Without it, the Bregman divergence would still be a useful algebraic object, but it would not provide the quantitative control needed for the final rate.

The theorem is also a useful interface point for later instantiations. To use the convergence theorem with a concrete mirror map, one must provide the strong convexity modulus. Once that is done, the general convergence argument no longer depends on the specific formula for φ.

Special Case Recovery

The most familiar mirror map is

`φ(x) = 1/2 ||x||^2.`

For this choice, `gradφ(x) = x`, and the Bregman divergence becomes

`D_φ(x, y) = 1/2 ||x - y||^2.`

The mirror update optimality condition then becomes the projected gradient descent optimality condition. In the unconstrained case it reduces further to

`x_{k+1} = x_k - α_k g_k.`

Thus projected gradient descent is the quadratic-mirror special case of the general theory formalised here.

This recovery is mathematically standard but not trivial to verify formally. One must show that the chosen φ has the required gradient, satisfies the strong convexity condition with the correct modulus, and makes the Bregman divergence definition simplify to squared Euclidean distance. The library's general theorems are stated so that such instantiations can be proved separately and then inherit the mirror descent convergence rate.

The negative entropy mirror map gives the other canonical special case:

`φ(x) = sum_i x_i log x_i.`

On the simplex, this produces the exponentiated-gradient or multiplicative-weights update. The point of the formalisation is that both examples are instances of the same Bregman proof, not separate convergence arguments.

5. Relation to Sibling Libraries

`mirror-descent-lean` is part of a broader Lean 4 formalisation programme for optimisation, dynamical systems, and learning theory.

`gradient-descent-lean` formalises gradient descent convergence and has DOI `10.5281/zenodo.20472996`. It is the closest sibling mathematically: projected gradient descent is recovered from mirror descent when the mirror map is `φ = 1/2 ||.||^2`. The present library moves from Euclidean squared distance to general Bregman geometry while preserving the same proof pattern of one-step descent followed by telescoping.

`nesterov-lean` formalises Nesterov accelerated gradient descent and has DOI `10.5281/zenodo.20474481`. Its headline rate is `O(1/k^2)` for smooth convex optimisation with momentum. By contrast, `mirror-descent-lean` proves the `O(1/sqrt k)` bounded-subgradient rate in a geometry adapted to the feasible set. The libraries therefore cover different axes of first-order optimisation: acceleration in Euclidean smooth settings versus non-Euclidean geometry for convex nonsmooth settings.

`contraction-lean` has DOI `10.5281/zenodo.20474762`. It formalises contraction-style reasoning, where convergence follows from a map shrinking distances. Mirror descent uses a different mechanism: Bregman divergence decreases in aggregate through a variational inequality and a telescoping potential.

`hopfield-lean` has DOI `10.5281/zenodo.20474169`. It formalises finite-state Hopfield energy descent and attractor convergence. Both libraries use energy-like functions to prove convergence, but the Hopfield argument is discrete and finite-state, while mirror descent is a convex optimisation argument over a real inner product space.

`kuramoto-lean` has DOI `10.5281/zenodo.20468619`. It formalises finite-N Kuramoto synchronisation dynamics and related Lyapunov-style identities. Like mirror descent, it shows how local differential or variational structure can imply global behaviour.

`lotka-volterra-lean` has DOI `10.5281/zenodo.20474669`. It contributes another dynamical-systems component to the same collection, focused on population dynamics and invariants rather than optimisation.

Together, these repositories form a set of machine-checked proof components for learning, optimisation, stability, and dynamics. `mirror-descent-lean` adds the Bregman-geometric first-order optimisation component.

6. AI Safety Significance

AI safety arguments often rely on claims about optimisation, convergence, monotonic improvement, stability, or bounded regret. These claims are easy to state informally but often depend on side conditions: convexity, differentiability, bounded gradients, positive step sizes, feasible iterates, strong convexity of a reference geometry, or exact update semantics. Formalisation forces those assumptions to be explicit.

Mirror descent is especially relevant because it is a template for optimisation under non-Euclidean geometry. Many safety-relevant systems are not naturally Euclidean. Probability distributions live on simplices, policies are constrained objects, and resource-allocation problems often have geometry better described by entropy or relative divergence than by squared norm. Bregman methods are one mathematical way to express optimisation that respects such structure.

The library does not certify any deployed AI system. Its value is foundational. It provides a checked proof that, under the stated hypotheses, a broad class of mirror descent procedures satisfies a quantitative averaged-regret guarantee. Future formal work can import the theorem rather than re-proving the convergence argument from scratch.

The proof also demonstrates a useful pattern for AI-safety formalisation: separate the generic optimisation theorem from the model-specific instantiation. The library proves the convergence theorem once for an abstract mirror map. A later project can instantiate the abstract hypotheses for a concrete geometry, a concrete feasible set, or a simplified learning rule.

This separation matters. In safety analysis, errors often arise when a general theorem is applied outside its domain of validity. A Lean library makes the boundary explicit. If an instantiation lacks strong convexity, a valid subgradient condition, or the required projection optimality property, the theorem cannot be applied without proving the missing hypothesis.

7. Conclusion

`mirror-descent-lean` provides a compact Lean 4 / Mathlib formalisation of mirror descent with Bregman divergences. It defines the Bregman divergence, first-order convexity conditions, mirror maps, single-step update data, and finite trajectories. It proves the non-negativity and equality characterisation of Bregman divergence, the three-point identity, the strong convexity lower bound, the mirror update estimates, the per-step regret bound, the telescoping sum, the constant-step averaged regret bound, and the final `O(1/sqrt K)` convergence rate.

The project is deliberately focused. It does not attempt to formalise every special mirror map, every equivalent definition of convexity, or every variant of mirror descent. Instead, it supplies the reusable convergence spine: Bregman geometry, update optimality, Young's inequality, strong convexity, telescoping, and rate optimisation.

As part of the broader family of Lean libraries for optimisation and dynamical systems, `mirror-descent-lean` adds a verified non-Euclidean first-order method. It can now serve as a foundation for future work on projected gradient descent as a special case, exponentiated gradient, multiplicative weights, online convex optimisation, and safety-relevant reasoning about learning dynamics.

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